Section 3.3 — Quadratic Functions — Answer Key

Math 103: College Algebra

1 Solving Quadratic Equations

1. $x = 2, 3$
2. $x = -3, 3$
3. $x = -2, 12$
4. $x = -\frac{5}{2}, \frac{1}{2}$
5. $x = -\frac{2}{3}, 6$
6. $x = -4, -2$

2 Graphing Quadratic Functions

For each of the following quadratic functions:

a. Find the x- and y-intercepts for the graph.

b. Using your x-intercepts, find the equation of the axis of symmetry and the coordinates of the vertex.

c. Create a table of x and y values that include at least the intercepts, the point with the same y-coordinate as the y-intercept, and two integer points on each side of the vertex.

d. Sketch the axis of symmetry as a dashed line, and plot the vertex. (Estimate fractions if necessary.)

e. Graph the function in the space provided, plotting all appropriate points from your table.

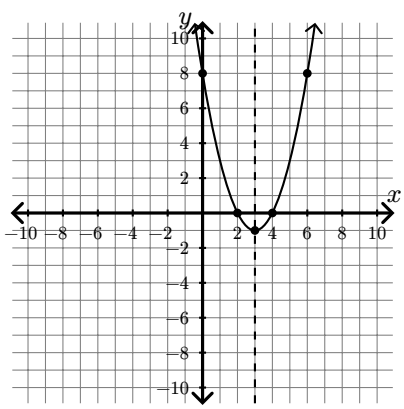
7.

a. y-intercept: $(0, 8)$
x-intercepts: $(2, 0)$,$(4, 0)$

b. Vertex: $(3, -1)$ Axis of Symmetry: $x = 3$

c.

x	0	2	3	4	6
y	8	0	-1	0	8



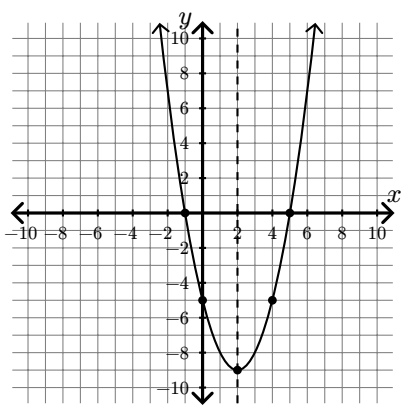
8.

a. y-intercept: $(0, -5)$
x-intercepts: $(-1, 0)$,$(5, 0)$

b. Vertex: $(2, -9)$ Axis of Symmetry: $x = 2$

c.

x	-1	0	2	5	4
y	0	-5	-9	0	-5



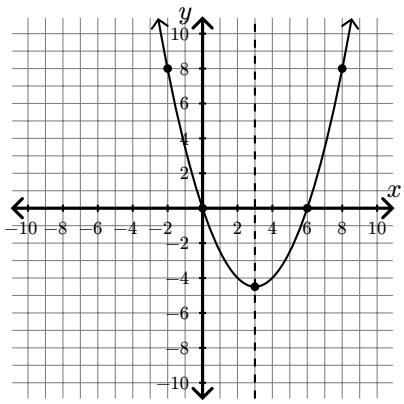
9.

a. y-intercept: $(0,0)$
x-intercepts: $(0,0)$,$(6,0)$

b. Vertex: $(3,-4\frac{1}{2})$ Axis of Symmetry: $x=3$

c.

x	-2	0	3	6	8
y	8	0	$-4\frac{1}{2}$	0	8



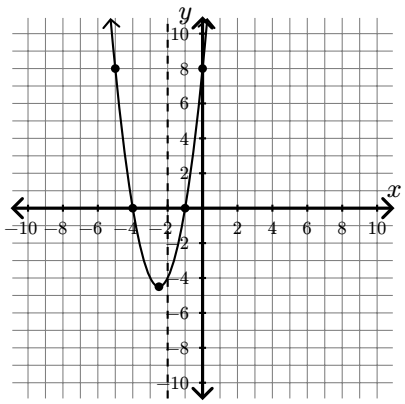
10.

a. y-intercept: $(0,8)$
x-intercepts: $(-4,0)$,$(-1,0)$

b. Vertex: $(-2\frac{1}{2},-4\frac{1}{2})$ Axis of Symmetry: $x=-2\frac{1}{2}$

c.

x	-5	-4	$-2\frac{1}{2}$	-1	0
y	8	0	$-4\frac{1}{2}$	0	8



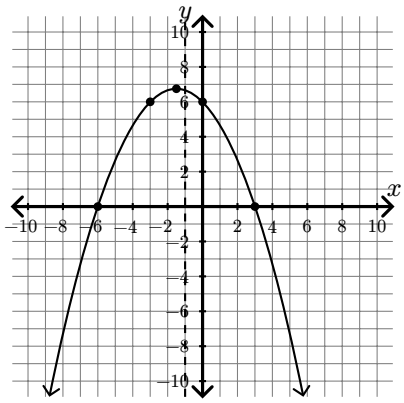
11.

a. y-intercept: $(0,6)$
x-intercepts: $(-6,0)$,$(3,0)$

b. Vertex: $(-1\frac{1}{2},6\frac{3}{4})$ Axis of Symmetry: $x=-1\frac{1}{2}$

c.

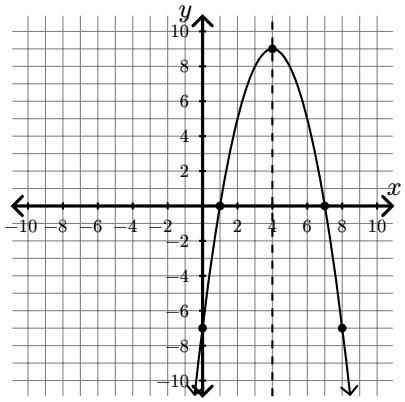
x	-6	-3	$-1\frac{1}{2}$	0	3
y	0	6	$6\frac{3}{4}$	6	0



12.

a. y-intercept: $(0,-7)$
x-intercepts: $(1,0)$,$(7,0)$

b. Vertex: $(4,9)$ Axis of Symmetry: $x=4$

c.

x	0	1	4	7	8
y	-7	0	9	0	-7